LIBBY — TARGET VALIDATION PATHS

Target validation paths — aml-mds-related-rr-tp53-aberrant-hla-pending-x7q2-rerun

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

Two workups decide whether this case's central question has an answer. Allele-level HLA-A*02:01 typing gates the HA-1/HA-2 TCR-T programs (TSC-100/101 via NCT05473910, the Bleakley HA-1 platform via NCT03326921, BSB-1001 via NCT06704152) and the WT1-C4 and PRAME-directed cells that share that allele; HLA-A*24:02, pulled in the same run, keeps a WT1-siTCR fallback alive if A*02:01 is negative. HA-1/HA-2 genotyping of both the patient and her HLA-identical sibling donor then establishes the recipient-positive / donor-negative mismatch those TCR-T products require. If HLA-A*02:01 and A*24:02 both type negative, this report has no within-scope HLA-restricted recommendation, and the next conversation about standard salvage care belongs to the treating team.

HLA-restricted immunotherapy

Two assays are essential and gate everything the case was submitted to surface. High-resolution HLA class I typing resolves the allele-level HLA-A*02:01 status that every HLA-restricted lever hangs on, with A*24:02 read in parallel so a negative A*02:01 does not close the door before the WT1-siTCR fallback is checked. Retrieve the allele-level ~2020 transplant typing report first and repeat high-resolution typing only if that record cannot be confirmed; turnaround is 3 to 7 days, with STAT available. HA-1/HA-2 minor-antigen genotyping is the second essential gate: HLA-identical siblings can still differ at these minor loci, so both the patient and the donor must be genotyped to establish the recipient HA-1(H)-positive / donor HA-1-negative mismatch that TSC-100/101 requires. It is an uncommon catalog test that runs 1 to 3 weeks as a specialized send-out and needs a fresh donor sample plus donor consent, so line those up early. Two high-priority reads add context rather than gate: standardized WT1 qPCR confirms the WT1-directed target and sets a quantitative MRD baseline, and a PRAME expression read (flow or RNA) establishes whether the PRAME-directed lever has a target present, since PRAME in AML is variable rather than universal.

TP53

TP53 is held aberrant pending confirmation, and the workup is built to resolve the 2019-mutant versus 2026-negative discrepancy without collapsing it to wild-type on a 2%-LOD panel that reads no copy number. Two essential steps carry it: retrieving the 2019 diagnostic variant identity and VAF from the original NGS report, and running a copy-number-capable heme NGS panel that reports TP53 at the variant level. A documented Y220C at either step is the specific handle for the Y220C-selective reactivator rezatapopt via NCT06616636; any other TP53 variant, or a true absence, removes that option. Two high-priority orthogonal reads corroborate the call from archival material at no new procedure: p53 protein IHC (an overexpression or complete-null pattern supports a functionally aberrant TP53) and 17p/TP53 deletion FISH (detects the copy loss the limited 2026 panel could not see).

CD33

Quantitative CD33 antigen-density flow (ABC/MESF) is the essential read here. It converts the qualitative CD33-positive call into the surface-density measure that predicts benefit from CD33-directed ADC therapy, since low density blunts gemtuzumab ozogamicin. Request the density quantification explicitly on the phenotyping specimen, because it is not a default readout. A medium-priority germline genotype of the CD33 rs12459419 splicing SNP refines the prediction: the CC genotype retains the target epitope and predicts better response, while a T allele blunts it. It does not gate on its own but tempers expectations alongside the density read.

CD123

Quantitative CD123 antigen-density flow (ABC/MESF) is the essential read for selecting among the CD123-directed agents (tagraxofusp, pivekimab sunirine). It can run on the same specimen and draw as the CD33 density request, and a dual CD33/CD123 density supports a combined-targeting strategy. Without it, CD123-directed cytoreduction ahead of a cellular or transplant consolidation is chosen blind to the predictive marker.

Second-transplant platform

Two high-priority confirmatory reads underpin the transplant plan. STR chimerism on flow-sorted CD34+ blasts confirms that the relapse arises from recipient-origin leukemia rather than a donor-derived or secondary leukemia; the sex-matched sibling donor rules out XY-FISH as a shortcut, and CD34+ sorting keeps normal donor hematopoiesis from masking the blast compartment. A full metaphase karyotype plus MECOM and KMT2A FISH at relapse confirms the adverse cytogenetic features (MECOM rearrangement, KMT2A amplification) that shape prognosis and trial eligibility.

Where to order these assays

Assay	Provider	Decision gated	Contact
High-resolution HLA class I typing (allele-level HLA-A02:01 and HLA-A24:02)	**HistoGenetics (preferred) (NGS high-resolution HLA typing)**	**HLA-A02:01/A24:02-restricted immunotherapy eligibility (TSC-100/101, WT1-C4, WT1-siTCR, PRAME-directed T cells)**	**test info · 300 Executive Blvd, Ossining, NY 10562 · (914) 762-6000**
High-resolution HLA class I typing (allele-level HLA-A02:01 and HLA-A24:02)	Versiti Diagnostic Laboratories	HLA-A02:01/A24:02-restricted immunotherapy eligibility (TSC-100/101, WT1-C4, WT1-siTCR, PRAME-directed T cells)	test info · 638 N 18th St, Milwaukee, WI 53233 · (800) 245-3117
High-resolution HLA class I typing (allele-level HLA-A02:01 and HLA-A24:02)	Labcorp	HLA-A02:01/A24:02-restricted immunotherapy eligibility (TSC-100/101, WT1-C4, WT1-siTCR, PRAME-directed T cells)	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
HA-1 / HA-2 minor histocompatibility antigen genotyping (recipient and HLA-identical sibling donor)	**Versiti Diagnostic Laboratories (preferred)**	**TSC-100/TSC-101 HA-1/HA-2 TCR-T (ALLOHA, NCT05473910); requires** **MiHA mismatch**	**test info · 638 N 18th St, Milwaukee, WI 53233 · (800) 245-3117**
HA-1 / HA-2 minor histocompatibility antigen genotyping (recipient and HLA-identical sibling donor)	Fred Hutchinson Cancer Center (Immunogenetics / MiHA research labs)	TSC-100/TSC-101 HA-1/HA-2 TCR-T (ALLOHA, NCT05473910); requires recipient-positive/donor-negative MiHA mismatch	test info · 1100 Fairview Ave N, Seattle, WA 98109 · (206) 667-5000

HA-1 / HA-2 minor histocompatibility antigen genotyping (recipient and HLA-identical sibling donor)	TScan Therapeutics (ALLOHA trial central testing)	TSC-100/TSC-101 HA-1/HA-2 TCR-T (ALLOHA, NCT05473910); requires recipient-positive/donor-negative MiHA mismatch	test info · 830 Winter St, Waltham, MA 02451
Retrieval of the 2019 diagnostic TP53 variant identity and VAF (original NGS report)	2019 testing laboratory (records request)	Rezatapopt (PC14586) via NCT06616636; Y220C-contingent	Request the annotated variant table (HGVS p. change and VAF) from the original 2019 testing laboratory
CNV-capable heme NGS panel with TP53 variant-level readout (Y220C determination)	**Foundation Medicine (preferred) (FoundationOne Heme)**	**Rezatapopt via NCT06616636 (Y220C-selective p53 reactivator)**	**test info · 150 Second St, Cambridge, MA 02141 · (888) 988-3639**
CNV-capable heme NGS panel with TP53 variant-level readout (Y220C determination)	Tempus (xT / xF)	Rezatapopt via NCT06616636 (Y220C-selective p53 reactivator)	test info · 600 W Chicago Ave, Suite 510, Chicago, IL 60654 · (800) 739-4137
CNV-capable heme NGS panel with TP53 variant-level readout (Y220C determination)	NeoGenomics Laboratories	Rezatapopt via NCT06616636 (Y220C-selective p53 reactivator)	test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907
Quantitative CD33 antigen-density flow cytometry (ABC/MESF)	**Hematologics, Inc. (preferred) (Differences from Normal (DfN) flow)**	**Gemtuzumab ozogamicin (CD33 ADC); dual CD33/CD123 targeting**	**test info · 3161 Elliott Ave, Suite 200, Seattle, WA 98121 · (800) 860-0745**
Quantitative CD33 antigen-density flow cytometry (ABC/MESF)	NeoGenomics Laboratories	Gemtuzumab ozogamicin (CD33 ADC); dual CD33/CD123 targeting	test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907
Quantitative CD33 antigen-density flow cytometry (ABC/MESF)	Labcorp	Gemtuzumab ozogamicin (CD33 ADC); dual CD33/CD123 targeting	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
Quantitative CD123 antigen-density flow cytometry (ABC/MESF)	**Hematologics, Inc. (preferred) (Differences from Normal (DfN) flow)**	**Tagraxofusp and pivekimab sunirine (CD123-directed)**	**test info · 3161 Elliott Ave, Suite 200, Seattle, WA 98121 · (800) 860-0745**
Quantitative CD123 antigen-density flow cytometry (ABC/MESF)	NeoGenomics Laboratories	Tagraxofusp and pivekimab sunirine (CD123-directed)	test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907
Quantitative CD123 antigen-density flow cytometry (ABC/MESF)	Labcorp	Tagraxofusp and pivekimab sunirine (CD123-directed)	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
Quantitative WT1 expression (ELN-standardized WT1 qPCR)	**Mayo Clinic Laboratories (preferred)**	**WT1-directed TCR-T (WT1-C4, WT1-siTCR) and WT1 MRD monitoring**	**test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710**
Quantitative WT1 expression (ELN-standardized WT1 qPCR)	Labcorp	WT1-directed TCR-T (WT1-C4, WT1-siTCR) and WT1 MRD monitoring	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
Quantitative WT1 expression (ELN-standardized WT1 qPCR)	ARUP Laboratories	WT1-directed TCR-T (WT1-C4, WT1-siTCR) and WT1 MRD monitoring	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · (800) 522-2787
PRAME expression on blasts (flow or qRT-PCR / RNA)	**NeoGenomics Laboratories (preferred) (PRAME IHC / RNA)**	**PRAME-directed T-cell / ImmTAC therapy (HLA-A*02:01-restricted)**	**test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907**

PRAME expression on blasts (flow or qRT-PCR / RNA)	Tempus (xR RNA)	PRAME-directed T-cell / ImmTAC therapy (HLA-A*02:01-restricted)	test info · 600 W Chicago Ave, Suite 510, Chicago, IL 60654 · (800) 739-4137
p53 protein immunohistochemistry (overexpression vs null pattern)	**Mayo Clinic Laboratories (preferred)**	**Corroborates TP53-aberrant call feeding the rezatapopt decision; not variant-level**	**test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710**
p53 protein immunohistochemistry (overexpression vs null pattern)	NeoGenomics Laboratories	Corroborates TP53-aberrant call feeding the rezatapopt decision; not variant-level	test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907
p53 protein immunohistochemistry (overexpression vs null pattern)	Labcorp	Corroborates TP53-aberrant call feeding the rezatapopt decision; not variant-level	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
17p / TP53 deletion FISH	**Mayo Clinic Laboratories (preferred)**	**TP53 copy-loss / multi-hit status; supports TP53-aberrant call and prognosis**	**test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710**
17p / TP53 deletion FISH	Labcorp	TP53 copy-loss / multi-hit status; supports TP53-aberrant call and prognosis	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
17p / TP53 deletion FISH	Quest Diagnostics	TP53 copy-loss / multi-hit status; supports TP53-aberrant call and prognosis	test info · 500 Plaza Dr, Secaucus, NJ 07094 · (866) 697-8378
STR chimerism on flow-sorted CD34+ blasts (recipient vs donor origin)	**Versiti Diagnostic Laboratories (preferred)**	**Clonal origin of relapse; informs HCT2 vs DLI vs alternative-donor strategy**	**test info · 638 N 18th St, Milwaukee, WI 53233 · (800) 245-3117**
STR chimerism on flow-sorted CD34+ blasts (recipient vs donor origin)	CareDx (AlloSeq HCT)	Clonal origin of relapse; informs HCT2 vs DLI vs alternative-donor strategy	test info · 8000 Marina Blvd, Suite 100, Brisbane, CA 94005 · (415) 287-2300
STR chimerism on flow-sorted CD34+ blasts (recipient vs donor origin)	Mayo Clinic Laboratories	Clonal origin of relapse; informs HCT2 vs DLI vs alternative-donor strategy	test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710
Full metaphase karyotype plus MECOM and KMT2A FISH at relapse	**Mayo Clinic Laboratories (preferred)**	**Adverse-cytogenetics confirmation (MECOM, KMT2A amplification) for prognosis and trial eligibility**	**test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710**
Full metaphase karyotype plus MECOM and KMT2A FISH at relapse	NeoGenomics Laboratories	Adverse-cytogenetics confirmation (MECOM, KMT2A amplification) for prognosis and trial eligibility	test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907
Full metaphase karyotype plus MECOM and KMT2A FISH at relapse	Labcorp	Adverse-cytogenetics confirmation (MECOM, KMT2A amplification) for prognosis and trial eligibility	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
CD33 rs12459419 splicing-SNP genotype	**Invivoscribe (preferred)**	**Refines gemtuzumab ozogamicin benefit prediction; not gating**	**test info · 10222 Barnes Canyon Rd, Building 1, San Diego, CA 92121 · (858) 224-6600**

CD33 rs12459419 splicing-SNP genotype	Labcorp	Refines gemtuzumab ozogamicin benefit prediction; not gating	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements
High-resolution HLA class I typing (allele-level HLA-A02:01 and HLA-A24:02)	Resolves the HLA-A02:01 (and HLA-A24:02) allele status that gates every HLA-restricted lever named for this case: HA-1/HA-2 TCR-T (TSC-100/101), WT1-C4, and PRAME-directed T cells are all presented on HLA-A02:01, with A24:02 opening a WT1-siTCR alternative if A*02:01 is negative. The allele-level result was generated at the ~2020 transplant workup but is not in the records on hand, so retrieve that report first and repeat high-resolution typing only if it cannot be confirmed at allele level. Without a confirmed allele call none of the HLA-restricted approaches, the stated reason for this run, can be scoped.	HistoGenetics (NGS high-resolution HLA typing) · test info · 300 Executive Blvd, Ossining, NY 10562 · (914) 762-6000	5-10 mL whole blood (EDTA); allele-level 2020 typing record acceptable if retrievable

HA-1 / HA-2 minor histocompatibility antigen genotyping (recipient and HLA-identical sibling donor)	Establishes the recipient HA-1(H)-positive / donor HA-1-negative (or HA-2) mismatch that HA-directed TCR-T requires; HLA-identical siblings can still differ at these minor histocompatibility loci, so both the patient and the sibling donor must be genotyped. Multiplex or allele-specific PCR resolves the HA-1H/HA-1R and HA-2 alleles from whole blood. Skipping the donor sample leaves TSC-100/101 eligibility undetermined even when HLA-A*02:01 is confirmed.	Versiti Diagnostic Laboratories · test info · 638 N 18th St, Milwaukee, WI 53233 · (800) 245-3117	5-10 mL whole blood (EDTA) from patient and sibling donor
Retrieval of the 2019 diagnostic TP53 variant identity and VAF (original NGS report)	Retrieves the exact amino-acid change and VAF from the 2019 diagnostic NGS so the 2019-mutant versus 2026-negative discrepancy can be adjudicated without down-calling to wild-type on a 2%-LOD panel that reads no copy number. A documented Y220C at diagnosis would directly support rezatapopt; any other TP53 variant would not. Acting on the limited 2026-negative result alone risks discarding a Y220C handle that the original report may already name.	2019 testing laboratory (records request); request the annotated variant table (HGVS p. change and VAF)	No new specimen; original 2019 NGS report and variant call file

CNV-capable heme NGS panel with TP53 variant-level readout (Y220C determination)	Determines at relapse whether the TP53 aberration is the Y220C variant using a copy-number-capable heme NGS panel with adequate sensitivity, rather than the 2%-LOD assay that reported negative without CNV calling. Y220C-positive disease is the specific handle for the Y220C-selective reactivator rezatapopt; non-Y220C or truly absent TP53 aberration removes that option. A single limited panel cannot exclude subclonal or copy-loss TP53 events, so a CNV-aware orthogonal read is needed before the target is called present or absent.	Foundation Medicine (FoundationOne Heme) · test info · 150 Second St, Cambridge, MA 02141 · (888) 988-3639	Fresh bone marrow aspirate at relapse (EDTA); high-blast peripheral blood acceptable
Quantitative CD33 antigen-density flow cytometry (ABC/MESF)	Converts the qualitative CD33-positive call into a quantitative antigen-density measure (ABC/MESF) that predicts benefit from CD33-directed ADC therapy, since low surface density blunts gemtuzumab ozogamicin activity. Request the density read on the same flow specimen used for phenotyping. Qualitative positivity alone does not establish whether the ADC handle is strong enough to act on.	Hematologics, Inc. (Differences from Normal (DfN) flow) · test info · 3161 Elliott Ave, Suite 200, Seattle, WA 98121 · (800) 860-0745	Fresh bone marrow aspirate or peripheral blood (heparin/EDTA), viable cells

Quantitative CD123 antigen-density flow cytometry (ABC/MESF)	Quantifies CD123 surface density on blasts to support selection among CD123-directed agents (tagraxofusp immunotoxin, pivekimab sunirine ADC), which target an antigen expressed on most AML blasts and leukemic stem cells. Request antigen-density flow on the phenotyping specimen. Without a density read, CD123-directed cytoreduction ahead of a cellular or transplant consolidation is chosen blind to the predictive marker.	Hematologics, Inc. (Differences from Normal (DfN) flow) · test info · 3161 Elliott Ave, Suite 200, Seattle, WA 98121 · (800) 860-0745	Fresh bone marrow aspirate or peripheral blood (heparin/EDTA), viable cells
Quantitative WT1 expression (ELN-standardized WT1 qPCR)	Confirms WT1 overexpression on the blasts, the target antigen for WT1-directed T-cell therapy (WT1-C4 on HLA-A02:01, WT1-siTCR on HLA-A24:02) and a validated marker to track measurable residual disease after any consolidation. Quantify by standardized WT1 qPCR on marrow or blood. Without a baseline WT1 level the WT1-directed lever cannot be confirmed as on-target and later MRD tracking has no reference point.	Mayo Clinic Laboratories · test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710	Fresh bone marrow or peripheral blood (EDTA)

PRAME expression on blasts (flow or qRT-PCR / RNA)	Measures PRAME expression on blasts to establish whether the PRAME-directed lever (PRAME TCR-T / ImmTAC, HLA-A02:01-restricted) has a target present, since PRAME expression in AML is variable rather than universal. Assess by flow or qRT-PCR on marrow or blood; commercial heme-specific PRAME assays are limited, so a send-out or academic RNA read may be needed. If PRAME is low or absent that pathway drops regardless of HLA-A02:01 status.	NeoGenomics Laboratories (PRAME IHC / RNA) · test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907	Fresh bone marrow or peripheral blood; FFPE marrow clot for IHC
p53 protein (overexpression vs null pattern)	Adds an orthogonal protein read of p53: a strong-overexpression or complete-null pattern corroborates a functionally aberrant TP53, while normal wild-type staining argues against it, helping adjudicate the 2019-versus-2026 NGS discrepancy. It runs on archival FFPE marrow, so it costs no new procedure. On its own IHC does not identify the specific variant, so it supports but does not replace variant-level NGS.	Mayo Clinic Laboratories · test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710	Archival FFPE bone marrow core or clot section
17p / TP53 deletion FISH	Detects 17p/TP53 copy loss that a 2%-LOD, no-CNV sequencing panel cannot see, addressing the most likely explanation (copy loss or subclonal loss) for the 2026 negative result. A deletion supports a multi-hit TP53 state with its adverse prognosis and reframes the discrepancy without collapsing it to wild-type. Omitting FISH leaves a plausible copy-loss mechanism unexamined.	Mayo Clinic Laboratories · test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710	Bone marrow aspirate (sodium heparin) or FFPE; archival acceptable

STR chimerism on flow-sorted CD34+ blasts (recipient vs donor origin)	Confirms that the relapse arises from recipient-origin leukemia (evolution of the original clone) rather than a donor-derived or secondary leukemia by running STR chimerism on flow-sorted CD34+ blasts; the sex-matched sibling donor rules out XY-FISH as a shortcut. Recipient-origin relapse with shared del(5q)/del(7q) supports a second-transplant plus donor-lymphocyte or HA-directed cellular strategy, whereas donor-derived leukemia would redirect the plan. Unsorted chimerism can be diluted by normal donor hematopoiesis and miss the blast compartment.	Versiti Diagnostic Laboratories · test info · 638 N 18th St, Milwaukee, WI 53233 · (800) 245-3117	Fresh bone marrow (EDTA) with CD34+ sorting; archived donor STR profile
Full metaphase karyotype plus MECOM and KMT2A FISH at relapse	Obtains a full metaphase karyotype plus targeted FISH for MECOM rearrangement and KMT2A amplification at relapse, confirming the adverse cytogenetic features that shape prognosis and trial eligibility. These lesions were reported on the 2026 panel and carry adverse risk under current AML frameworks. Without a complete relapse karyotype, risk stratification and several trial screens rest on an incomplete cytogenetic picture.	Mayo Clinic Laboratories · test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710	Fresh bone marrow aspirate (sodium heparin), 2-3 mL

CD33 rs12459419 splicing-SNP genotype	Genotypes the CD33 rs12459419 splicing SNP: the CC genotype retains the IgV epitope and predicts substantially better gemtuzumab ozogamicin response, while the T allele reduces surface CD33 and blunts benefit. It refines whether a CD33-ADC lever is worth pursuing alongside the antigen-density read. Not gating on its own, but a CT/TT result tempers expectations for gemtuzumab ozogamicin.	Invivoscribe · test info · 10222 Barnes Canyon Rd, Building 1, San Diego, CA 92121 · (858) 224-6600	Whole blood or bone marrow (germline-inferred; buccal swab acceptable)

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.